

97770**BCA – NEP 4 Year 4th Semester****Examination, May-2026****COMPUTER GRAPHICS****Paper : 24BCA404DS04***Time allowed : 3 Hours]**[Maximum marks : 50*

Note : Question 1 is compulsory having five parts and each part is of 2 marks total of 10 marks and the remaining questions are of 10 marks. And attempt one question from each section.

- 1.** Write a short note on the following : **5×2=10**
- (a) Define pixel.
 - (b) List the steps in scan conversion.
 - (c) What is the use of transformation matrices?
 - (d) What is a clipping window?
 - (e) Define spline and its importance.

Unit-I

- 2.** Discuss interactive and non-interactive graphics with applications. **10**
- 3.** Explain video display devices and input devices in graphics systems. **10**

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Unit-II

4. Implement Bresenham's line algorithm for a line form (2, 3) to (10, 8). 10
5. Describe mid-point circle drawing algorithm. 10

Unit-III

6. Prove that two successive rotations are additive in 2D transformation. 10
7. Explain 2D reflection and shearing with example matrices. 10

Unit-IV

8. Write a short note on Polygon clipping. 10
9. Differentiate between viewport and window. Explain mapping between them. 10

